LLMs/ChatGPT and a Generative Future

Keeping the Human in the Loop

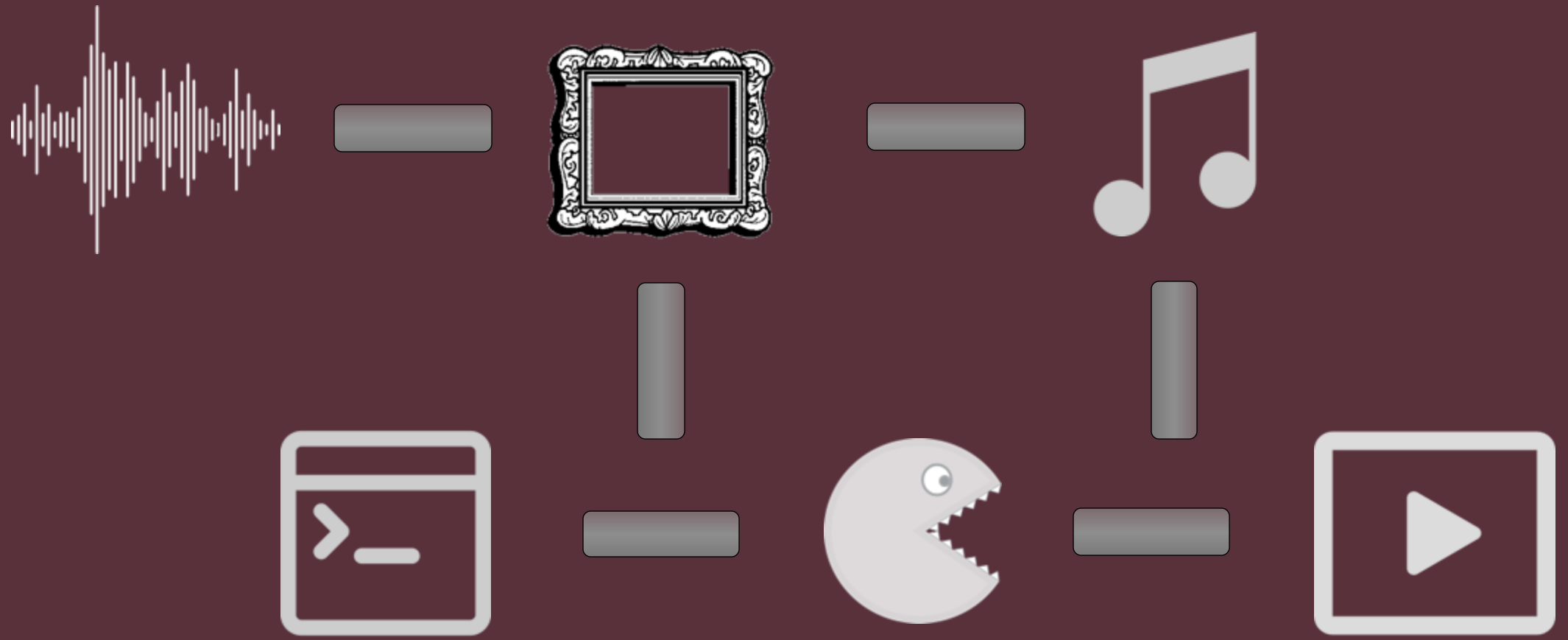
Slides in here can be referenced as Slides Week 1B: 1 etc.

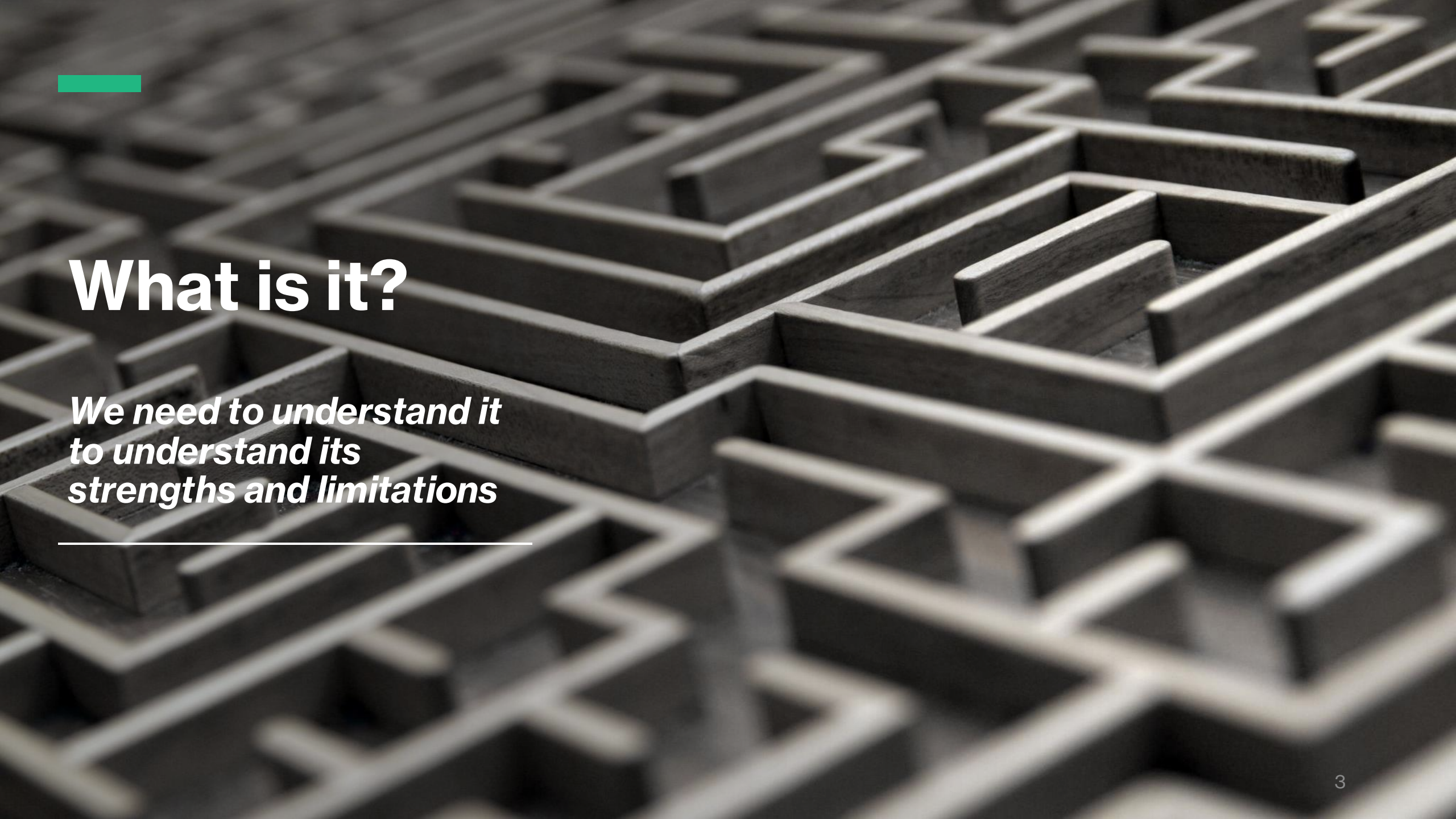
Robert Ross, TU Dublin

Cognitive Systems

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The Generative AI Landscape

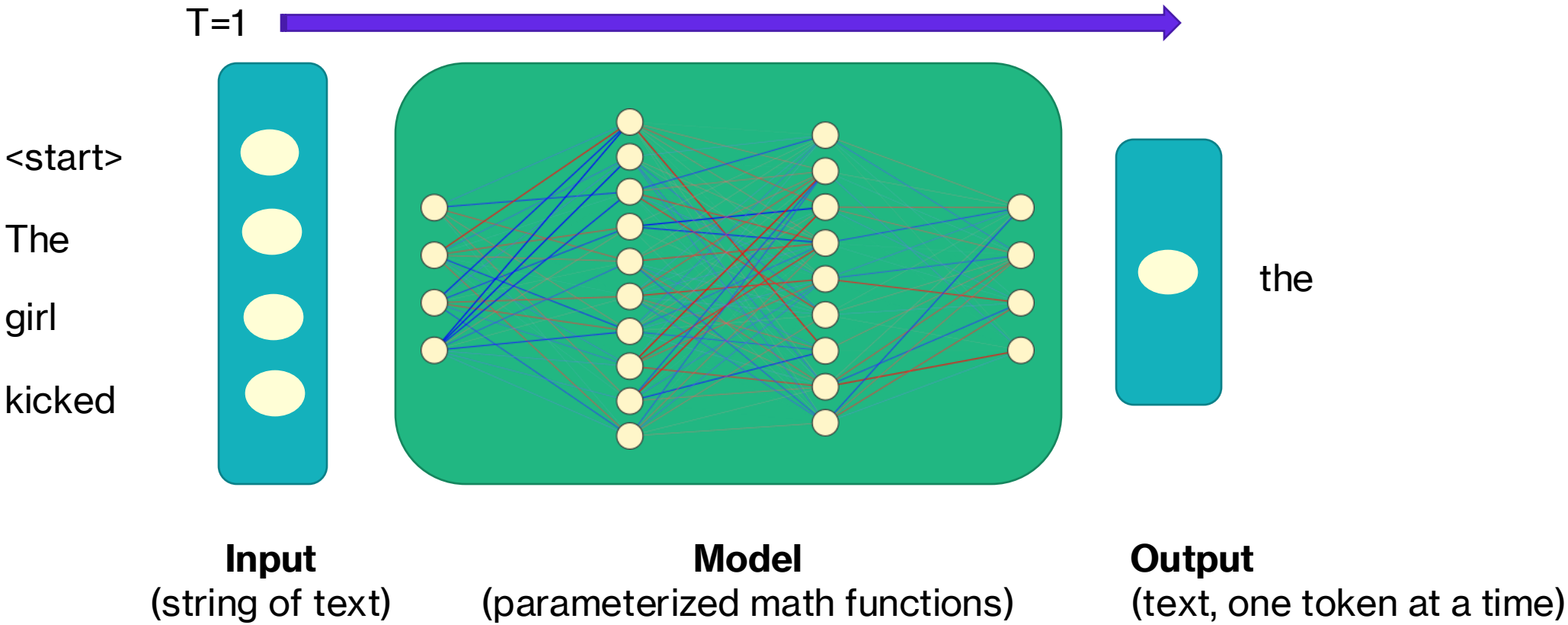




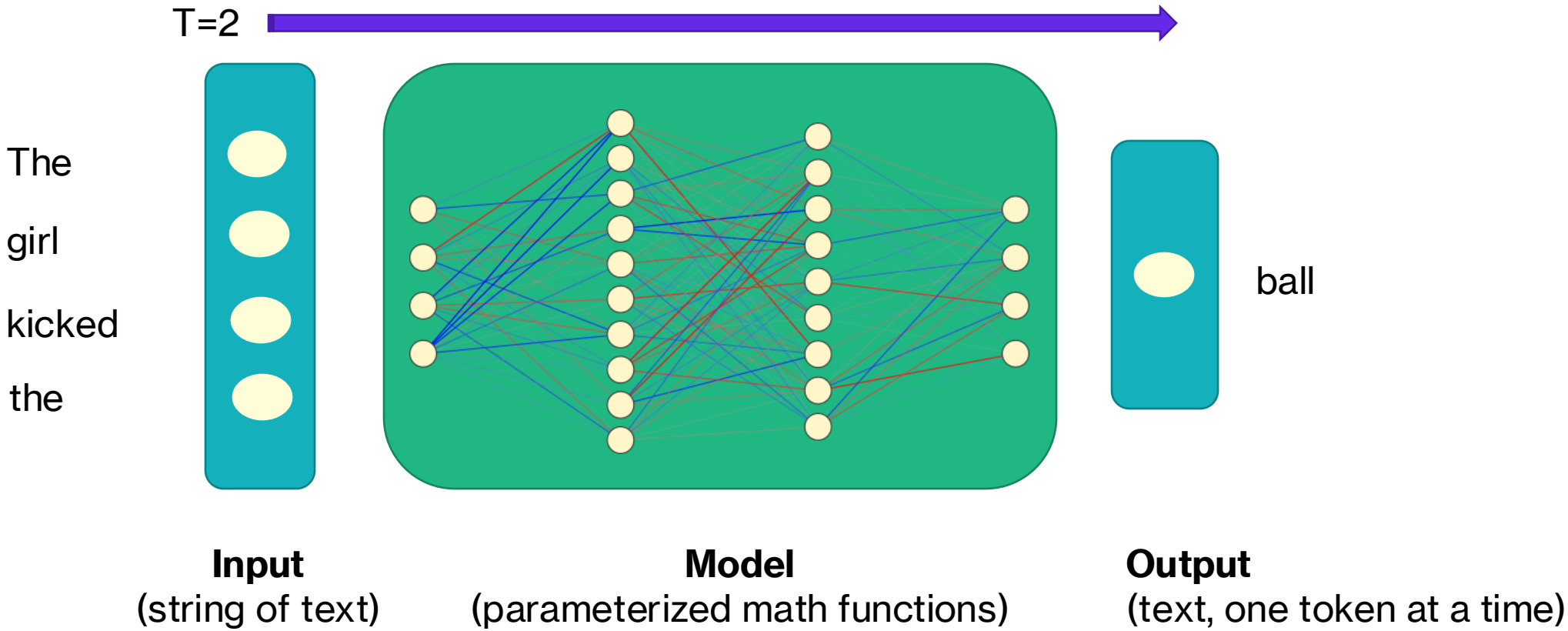
What is it?

We need to understand it to understand its strengths and limitations

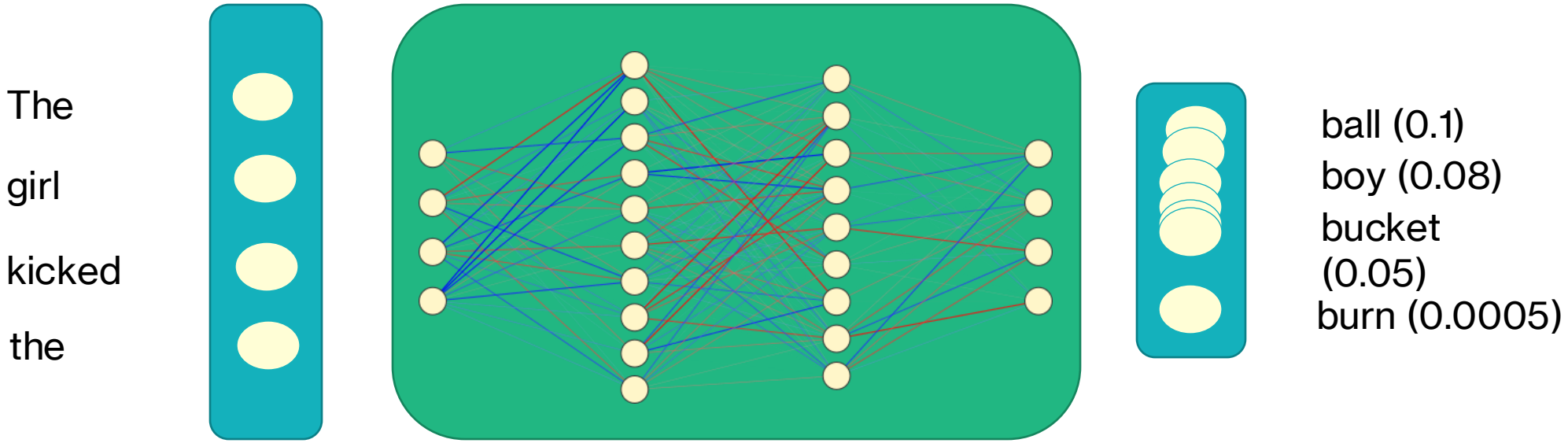
A neural language model



A neural language model



A neural language model (detail)

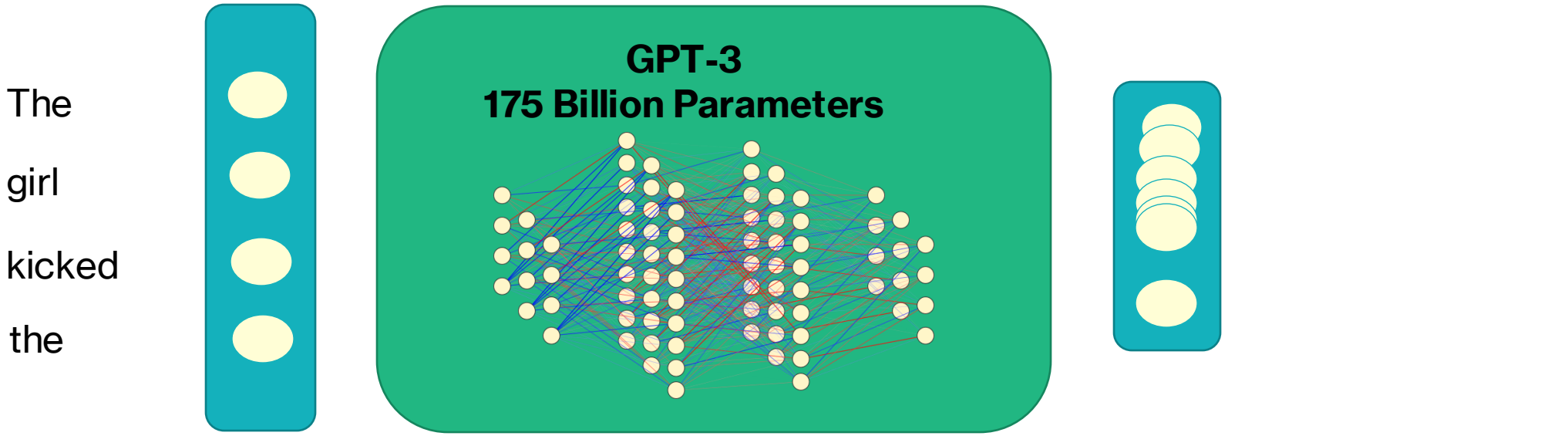


Input
(string of text)

Model
(parameterized math functions)

Output
(text, one token at a time)

A large language model



Input
(string of text)

Model
(parameterized math functions)

Output
(text, one token at a time)

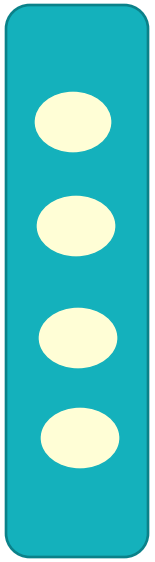
A conversational language model

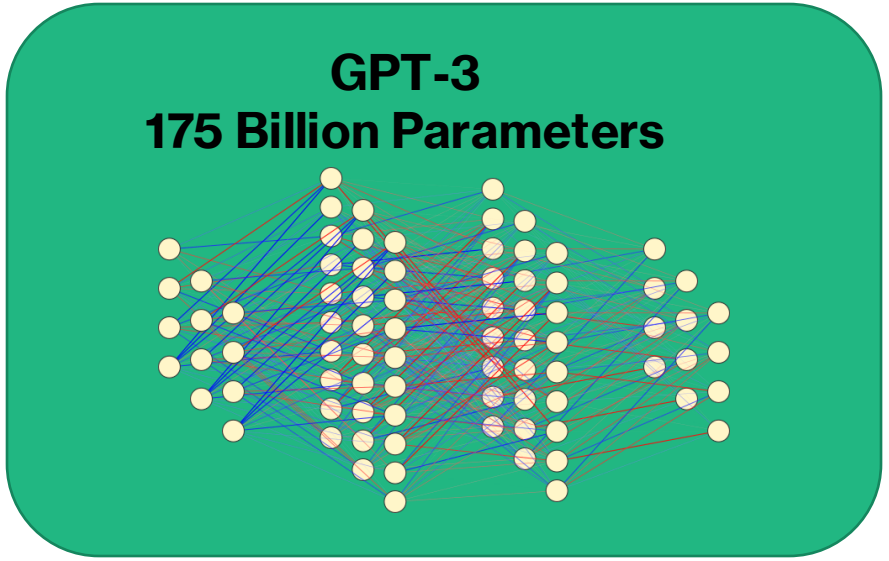
What

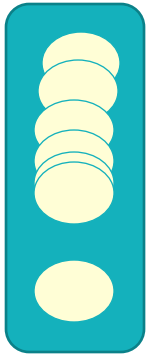
is

your

name?







assistant

is

my

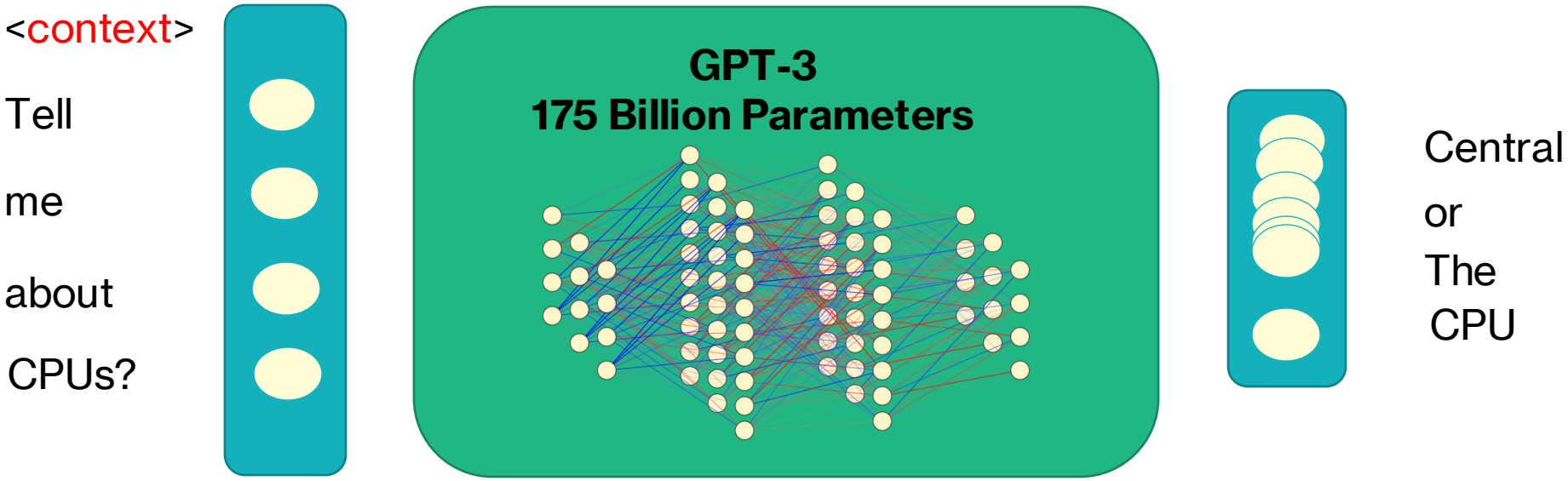
name

Input
(string of text)

Model
(parameterized math functions)

Output
(text, one token at a time)

A ChatGPT like language model



Input
(string of text)

Model
(parameterized math functions)

Output
(text, one token at a time)

Aside: MODERN LLM ORIGIN:

The Transformer Architecture

Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Niki Parmar* Google Research nikip@google.com

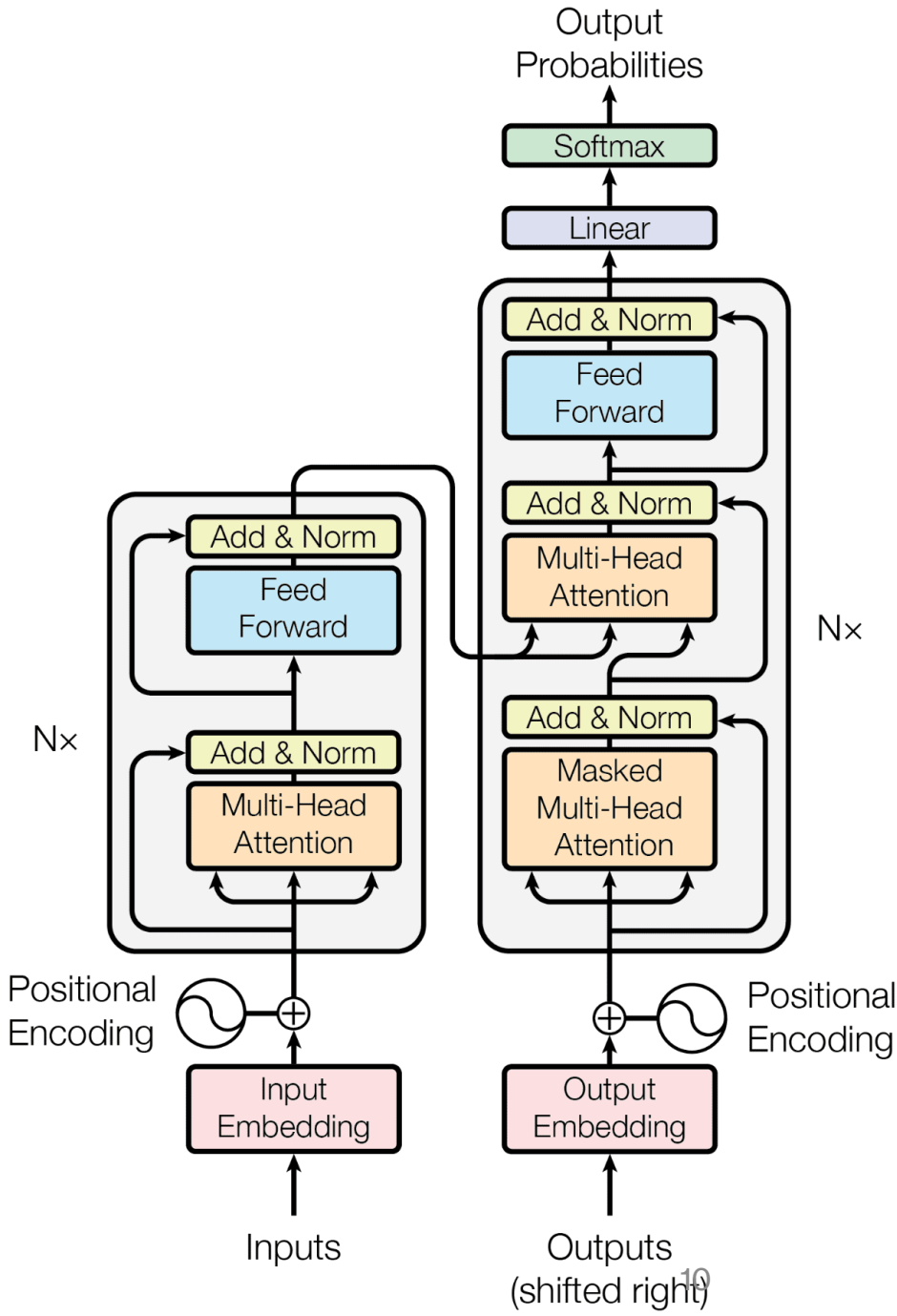
Jakob Uszkoreit* Google Research usz@google.com

Llion Jones* Google Research llion@google.com

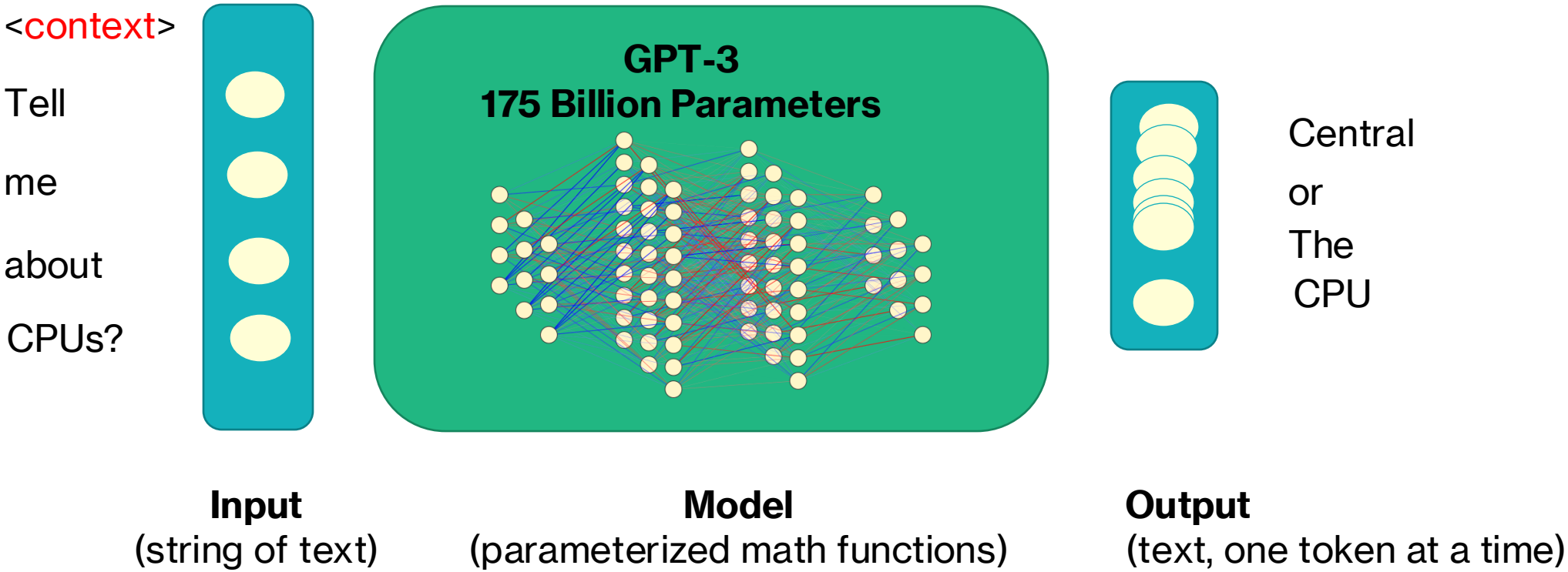
Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

Illia Polosukhin* ‡ illia.polosukhin@gmail.com



A ChatGPT like language model



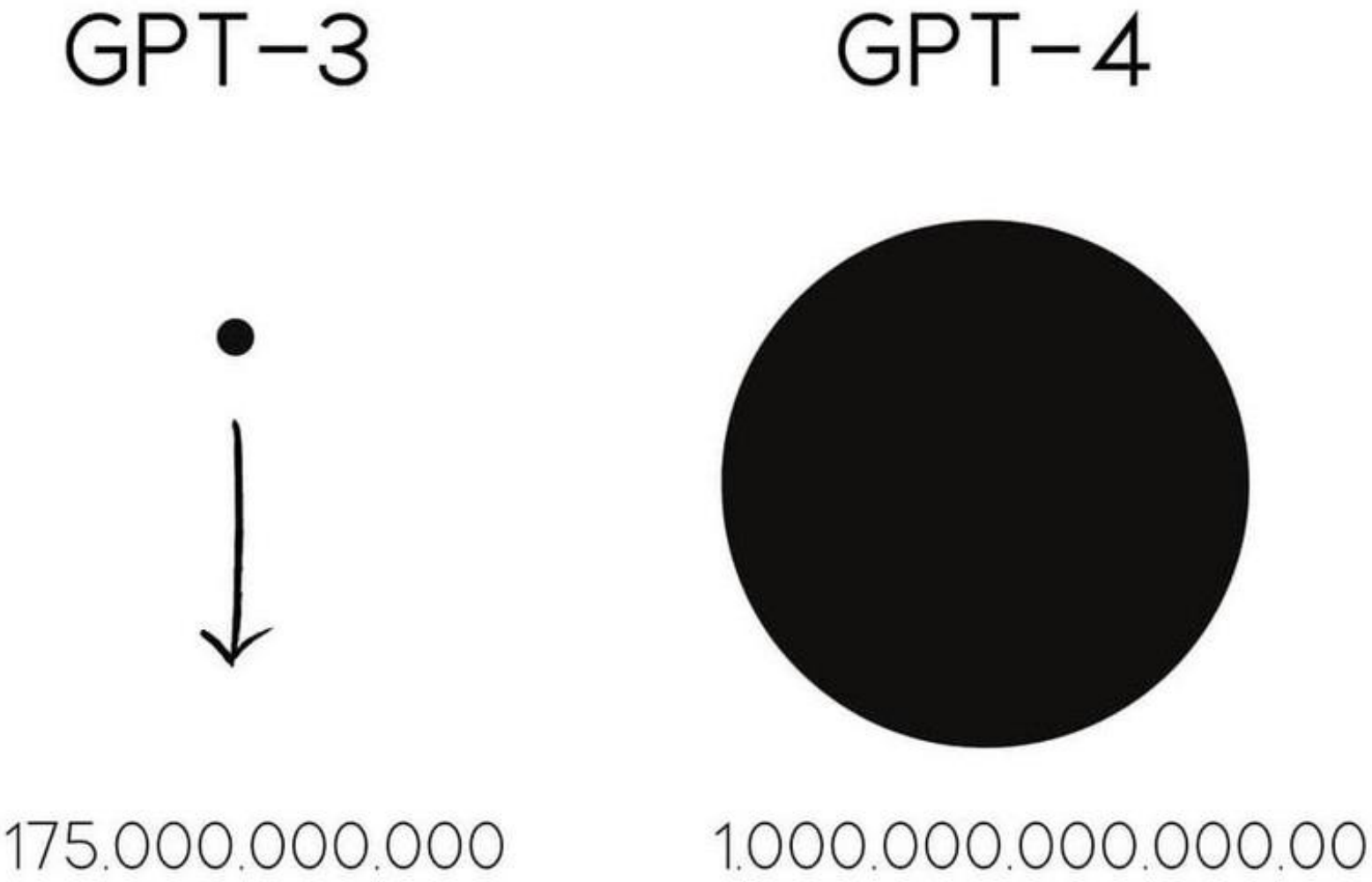
Putting some ‘safety’ in place.. .



Pre-Filter
User Input

LLM Core

Post-Filter
System Output

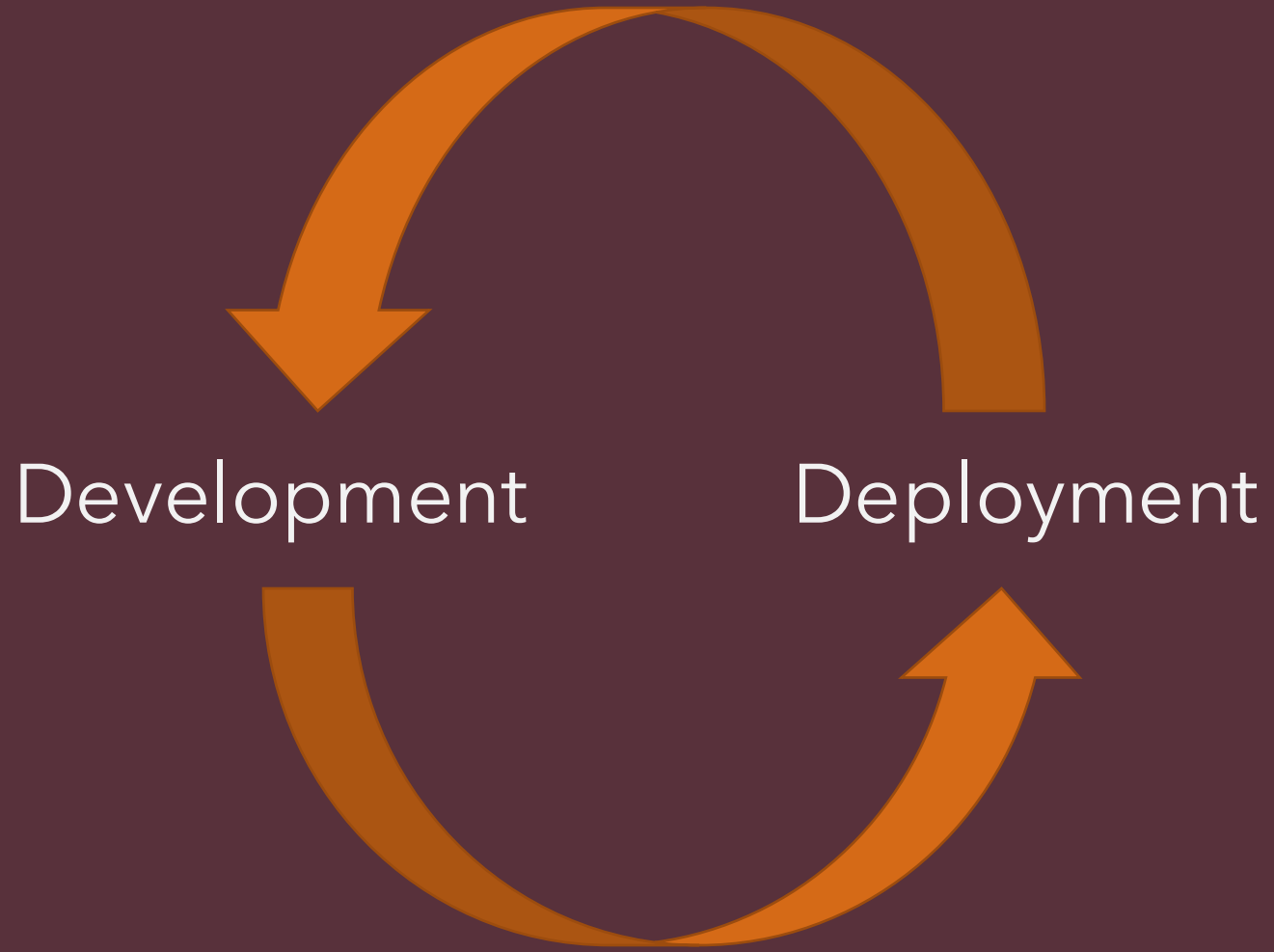


https://www.researchgate.net/figure/The-comparison-between-GPT-3-and-GPT-4-based-on-the-number-of-parameters-used-in-their_fig3_369977556

Number of Parameters source of debate.


terminally onλine engi...
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Keeping the Human in the Loop

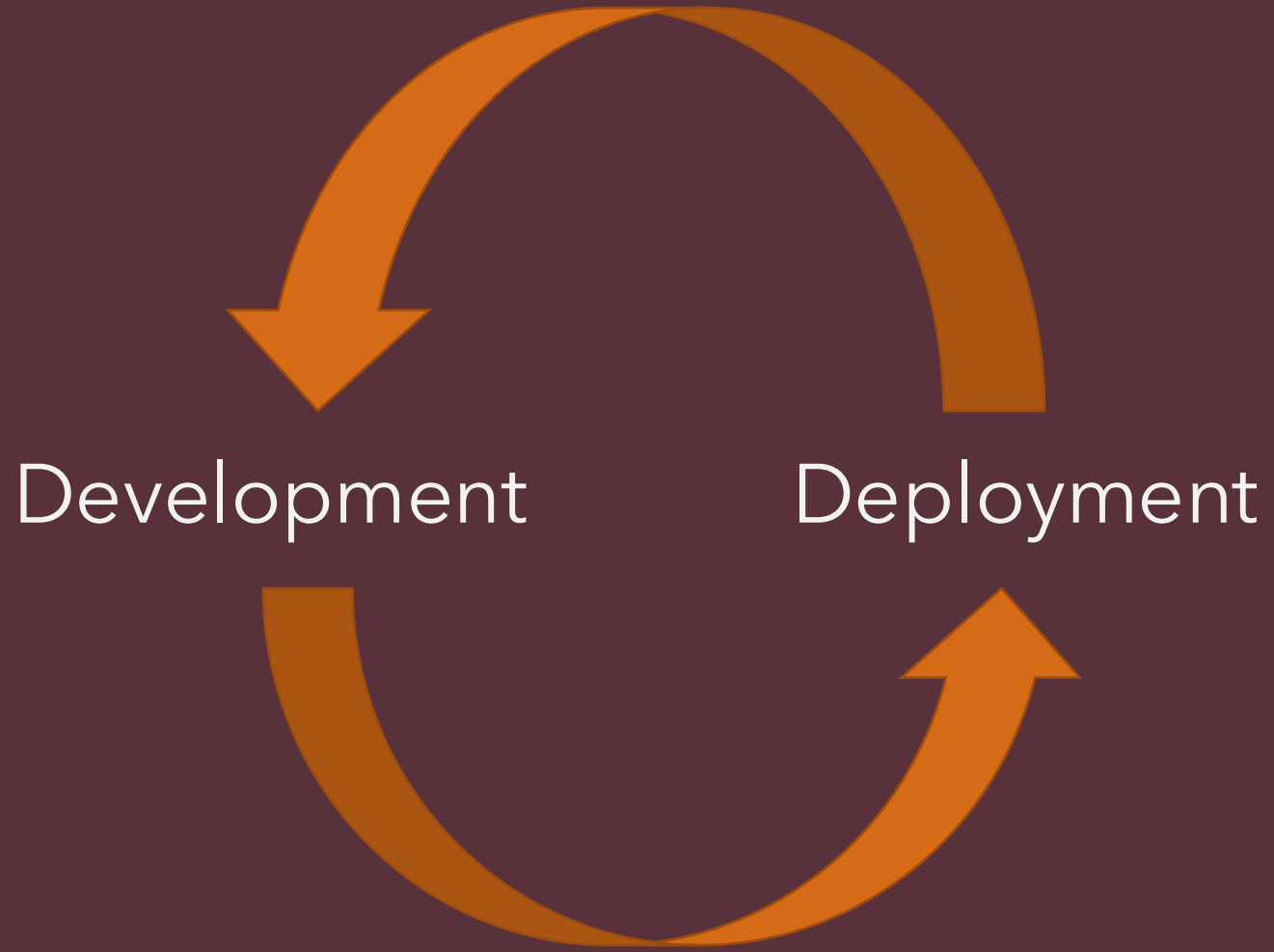


Ethics & Governance

Personalisation

Training

Keeping the Human in the Loop

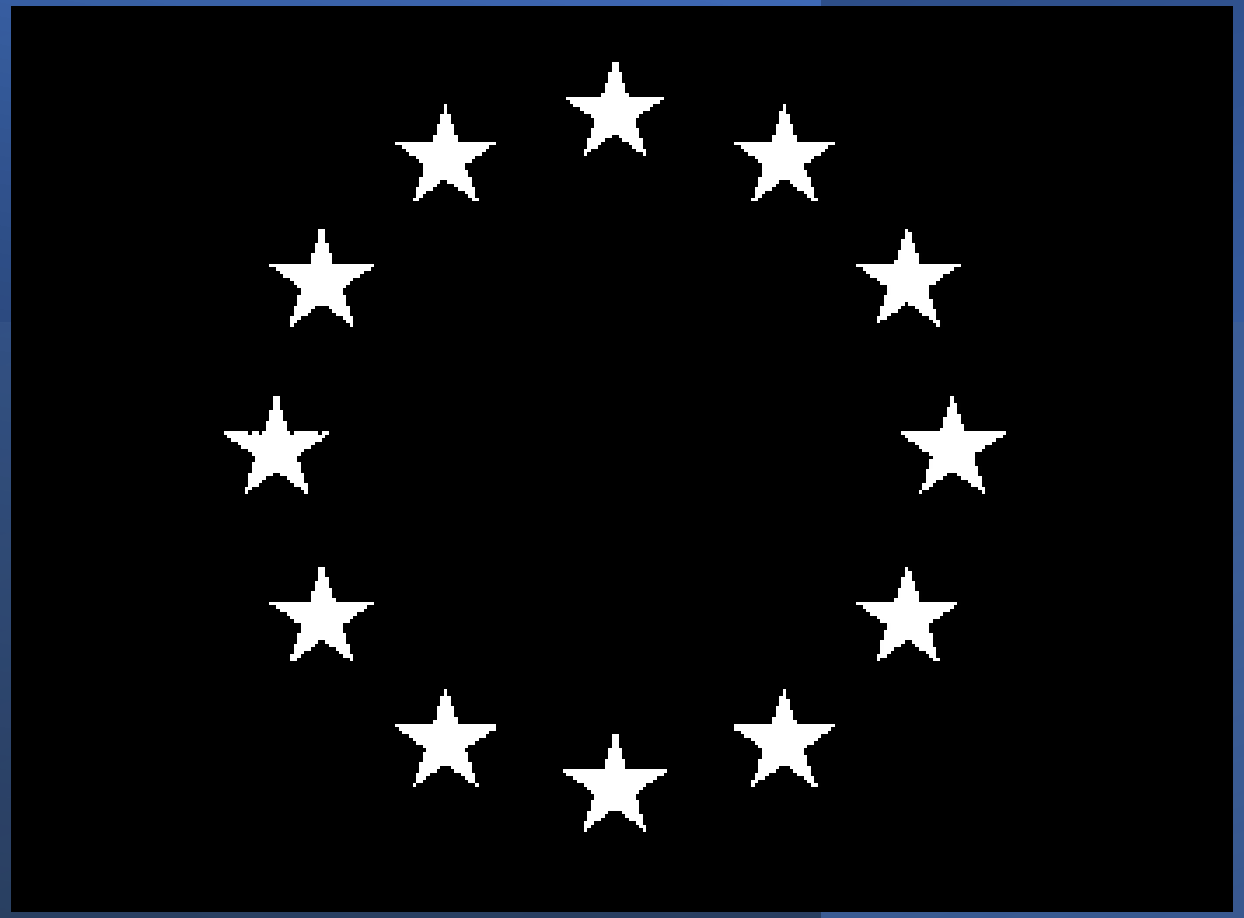


Ethics & Governance

Personalisation

Training

Ethics & Governance



The AI Act

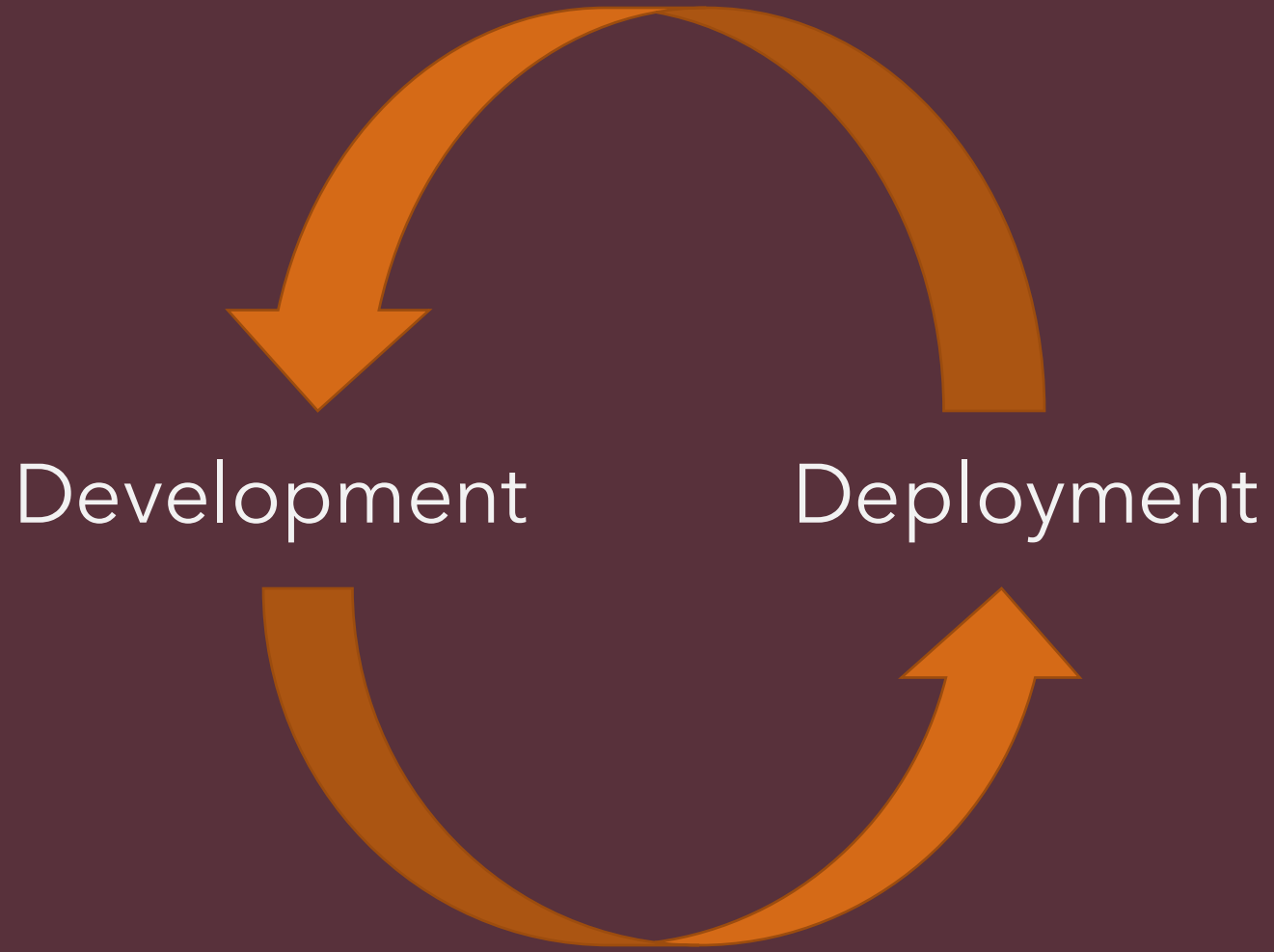
The Digital Services Act

Coimisiún na Meán



Niamh Hodnett
Online Safety Commissioner

Keeping the Human in the Loop

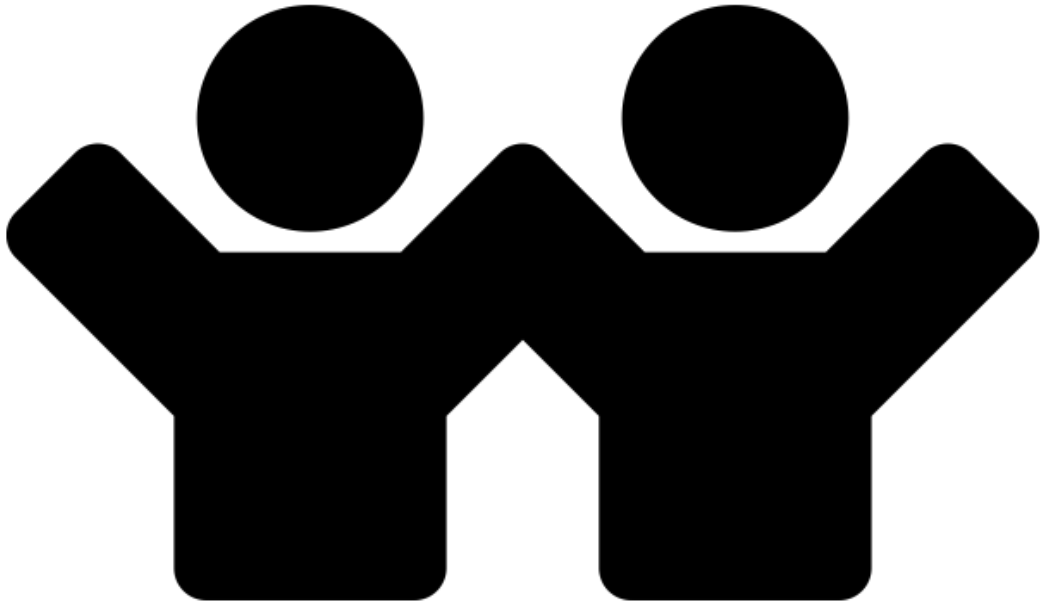


Ethics & Governance

Personalisation

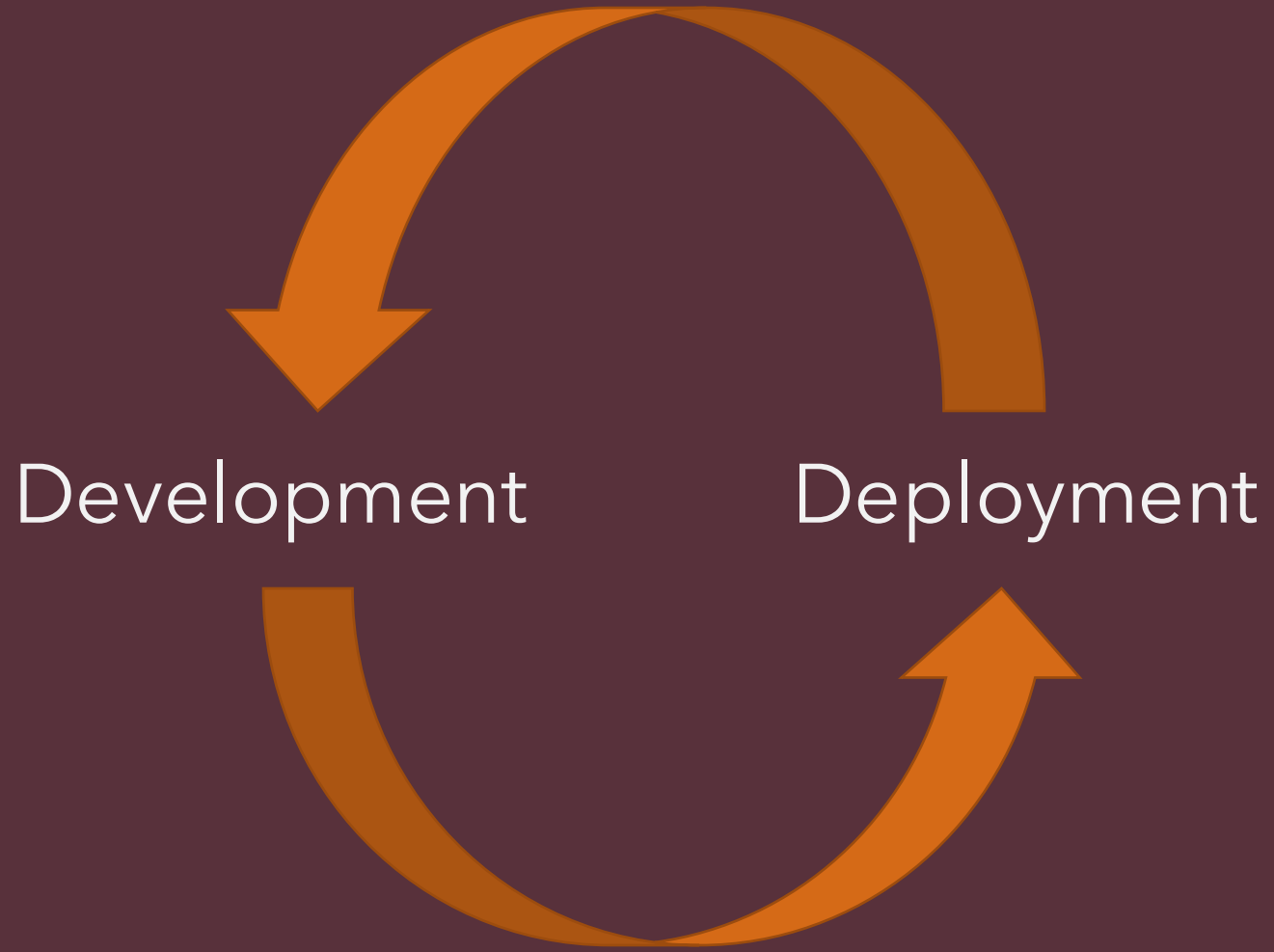
Training





What role should conversational tech play?

Keeping the Human in the Loop

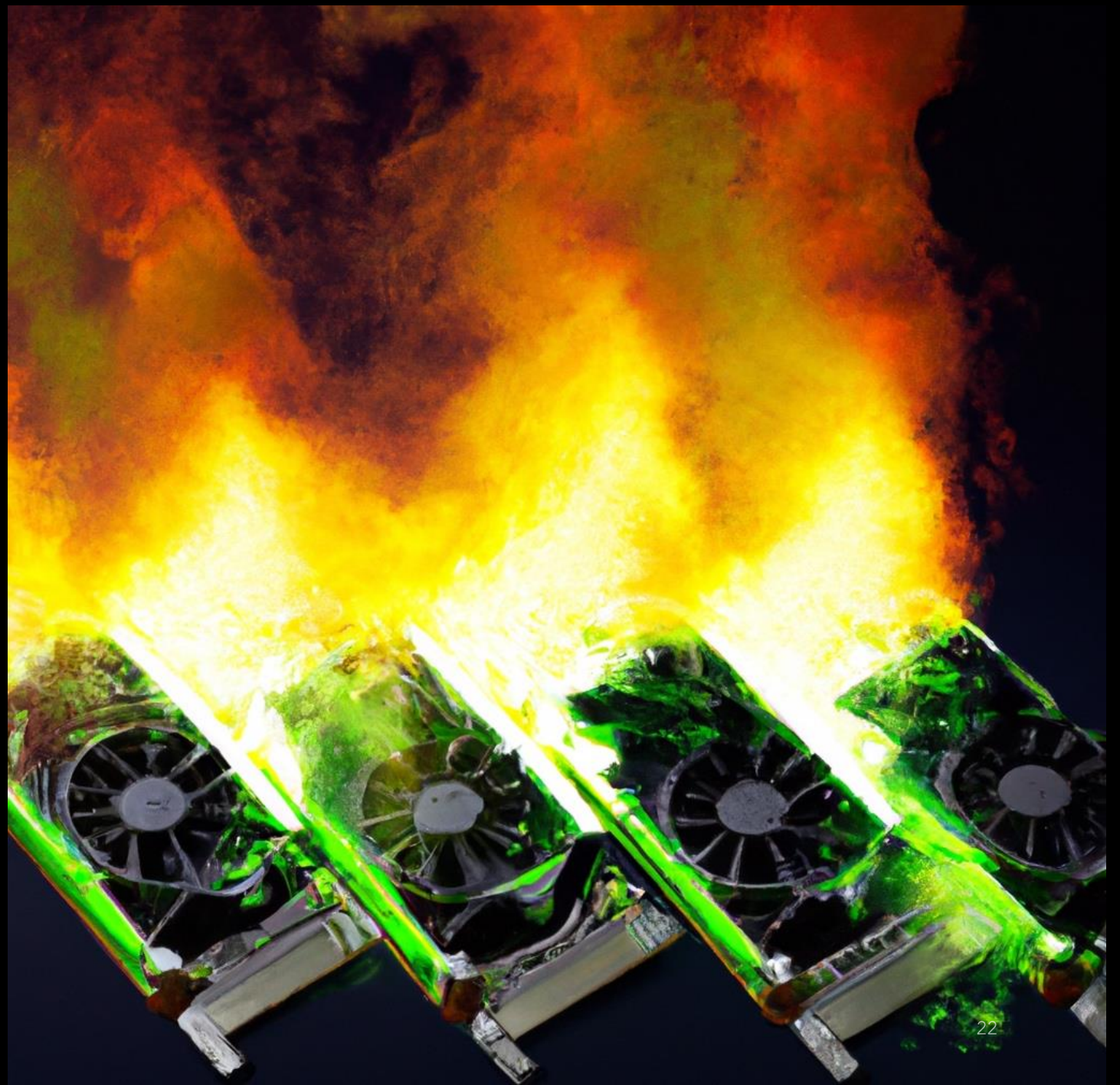


Personalisation

Ethics & Governance

Training

The Big Data Assumption



OpenAI ChatGPT

- Code Writing
- Code Debugging
- Question Answering
- Write (ok?) Poetry
- Synthesising (mediocre) stories
- ….

- No Built-in Embodiment
- Limited Personalisation
- Current Age of Data
- Lack of Naturalness
- A Generative Blur
- ….

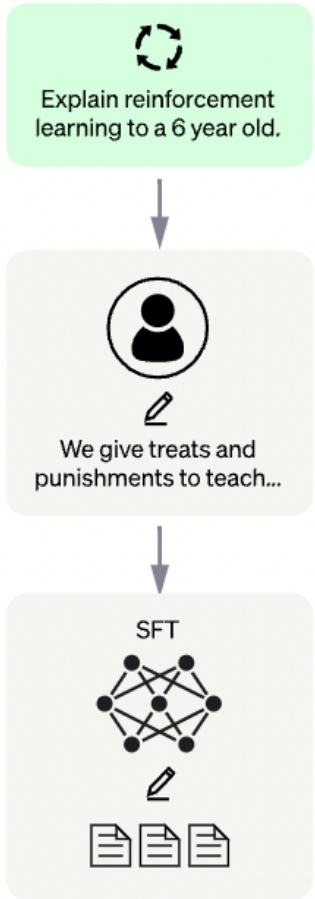
Step 1

Collect demonstration data and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3.5 with supervised learning.



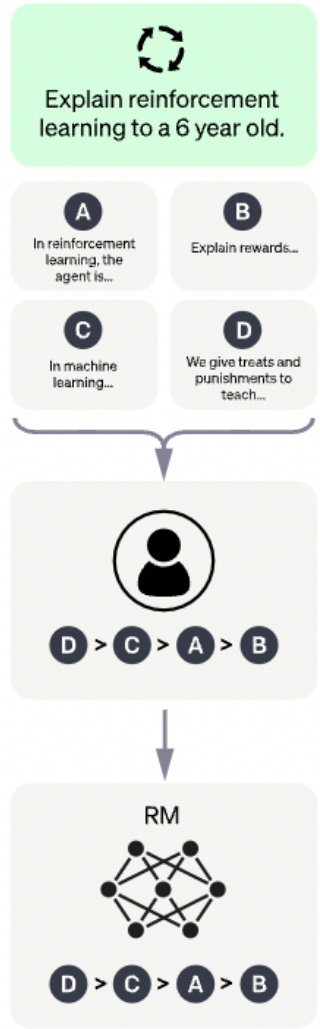
Step 2

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

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Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

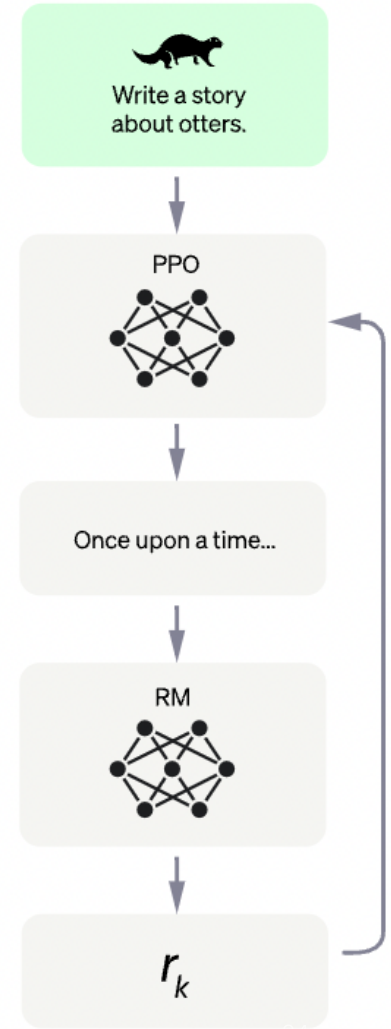
A new prompt is sampled from the dataset.

The PPO model is initialized from the supervised policy.

The policy generates an output.

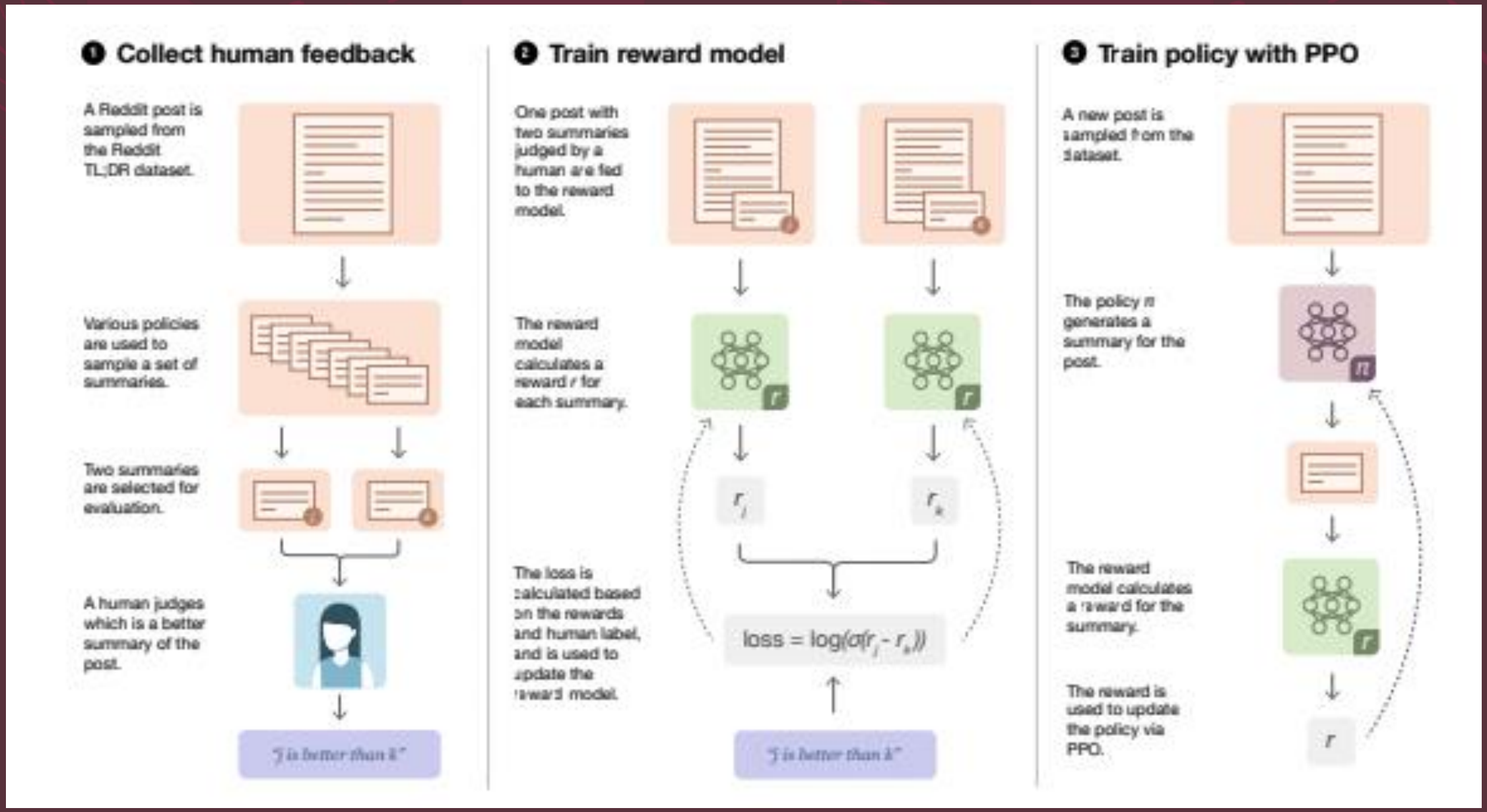
The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.

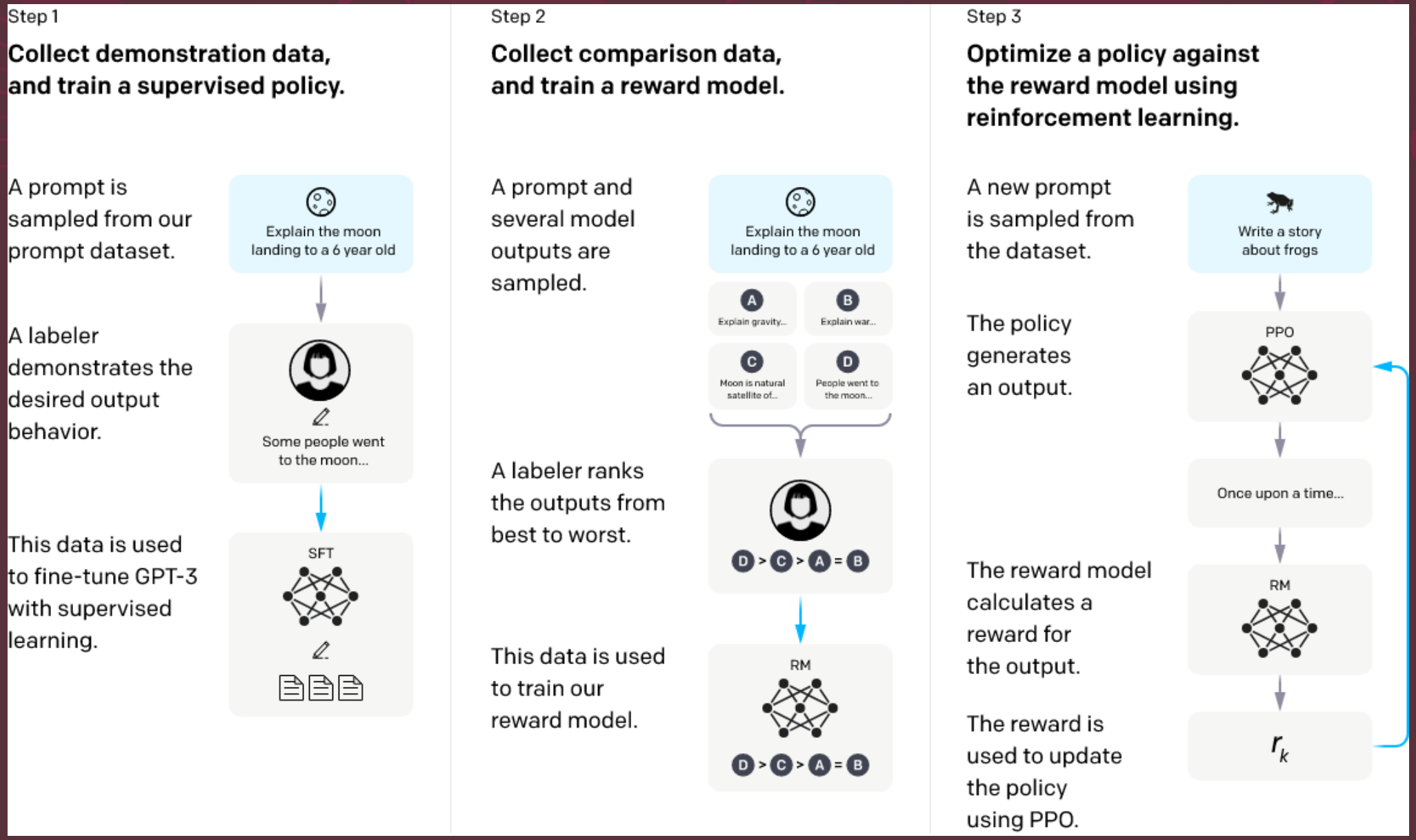




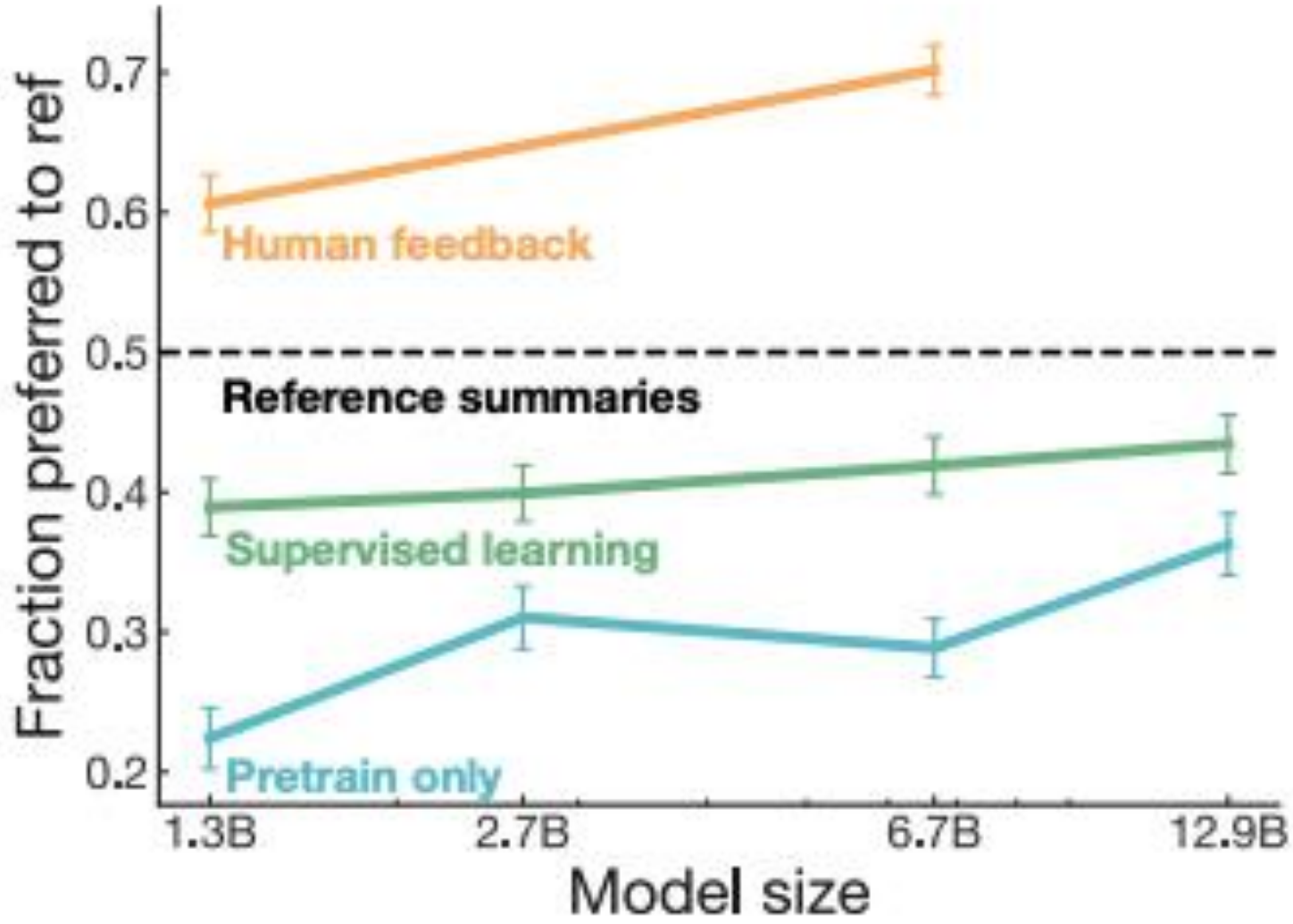
OpenAI



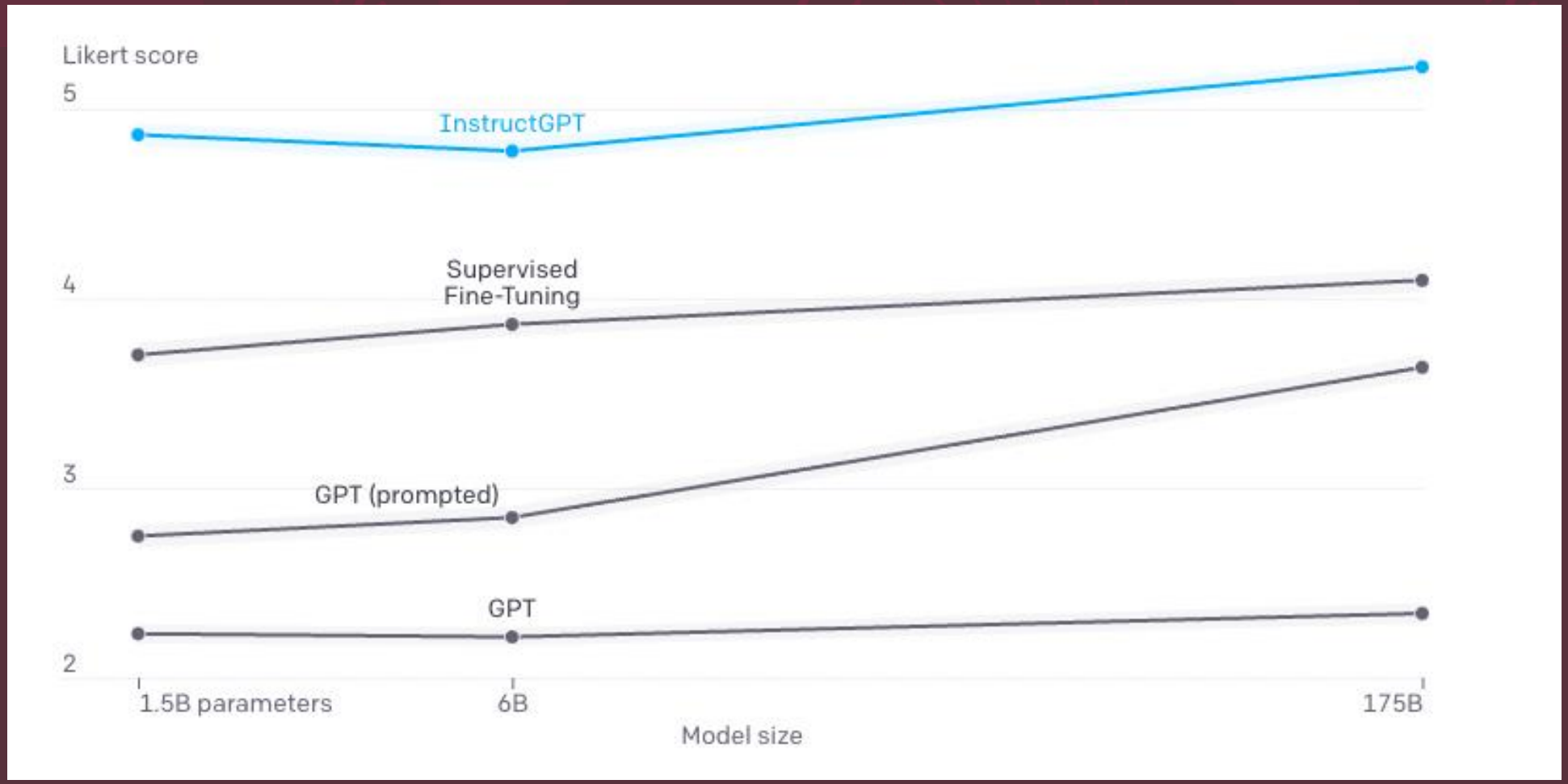
Learning to summarize from human feedback - Training



Aligning Language Models to Follow Instructions - Training



Learning to summarize from human feedback - Illustrative



Aligning Language Models to Follow Instructions - Illustrative

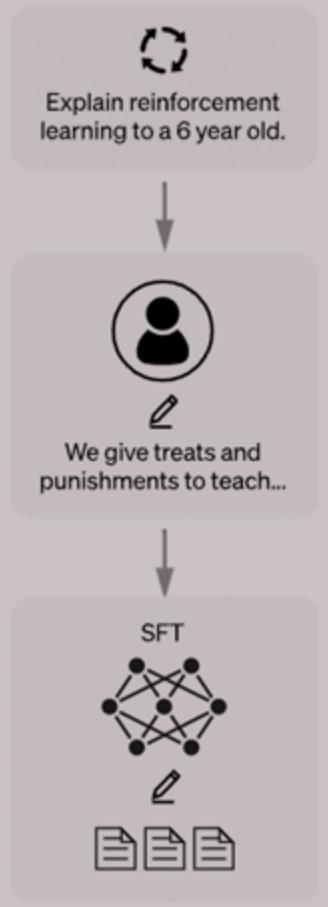
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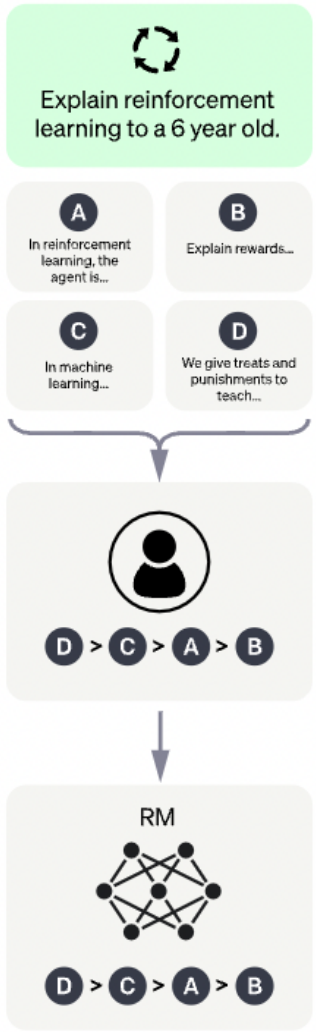
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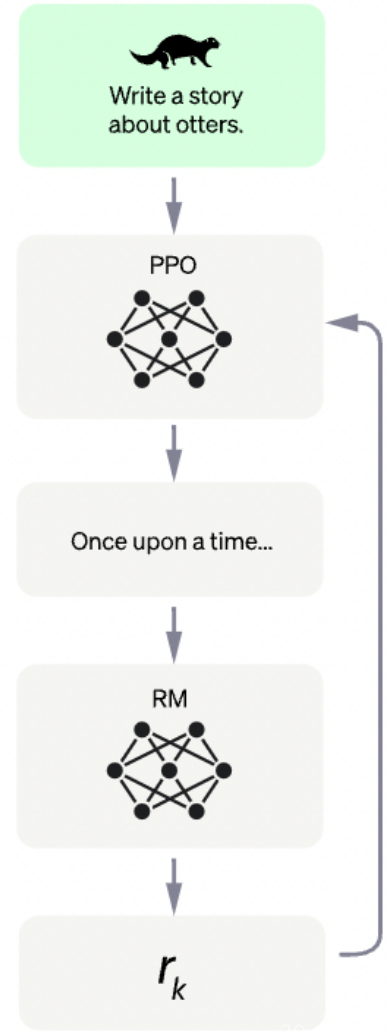
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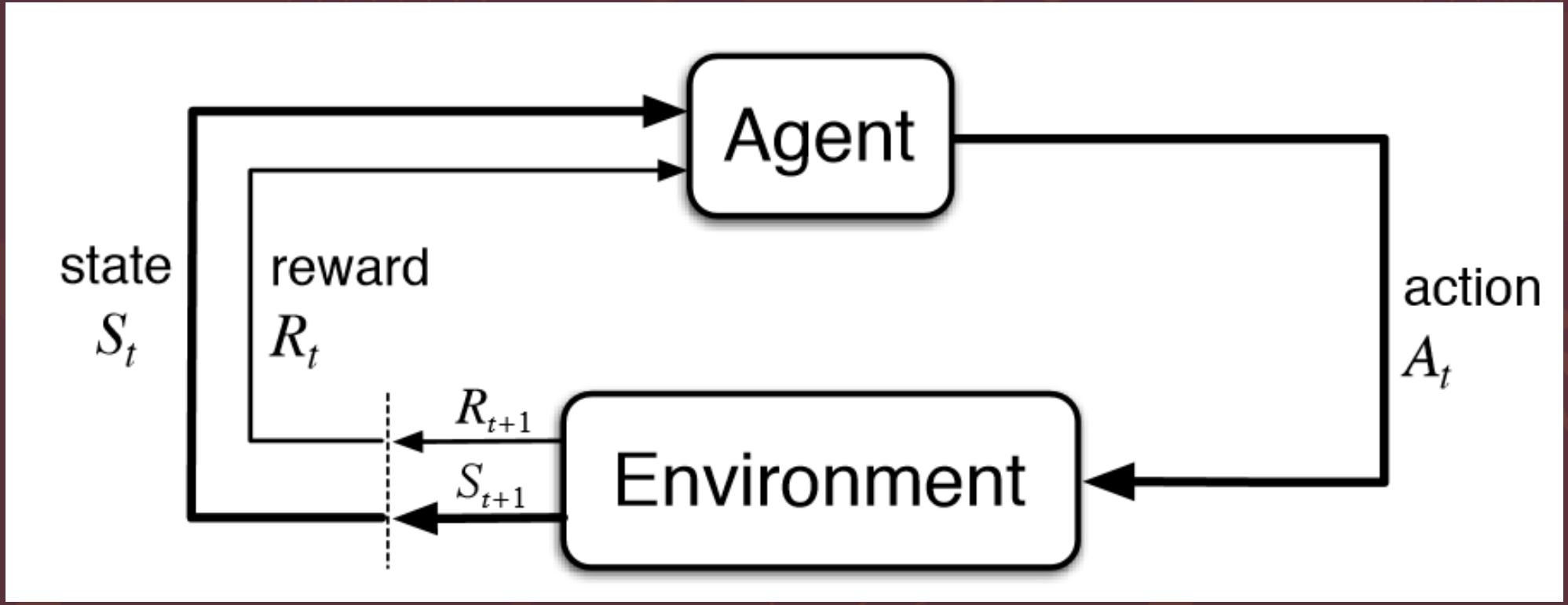
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Reinforcement Learning



Illustrating Reinforcement Learning from Human Feedback (RLHF)

Published December 9, 2022.

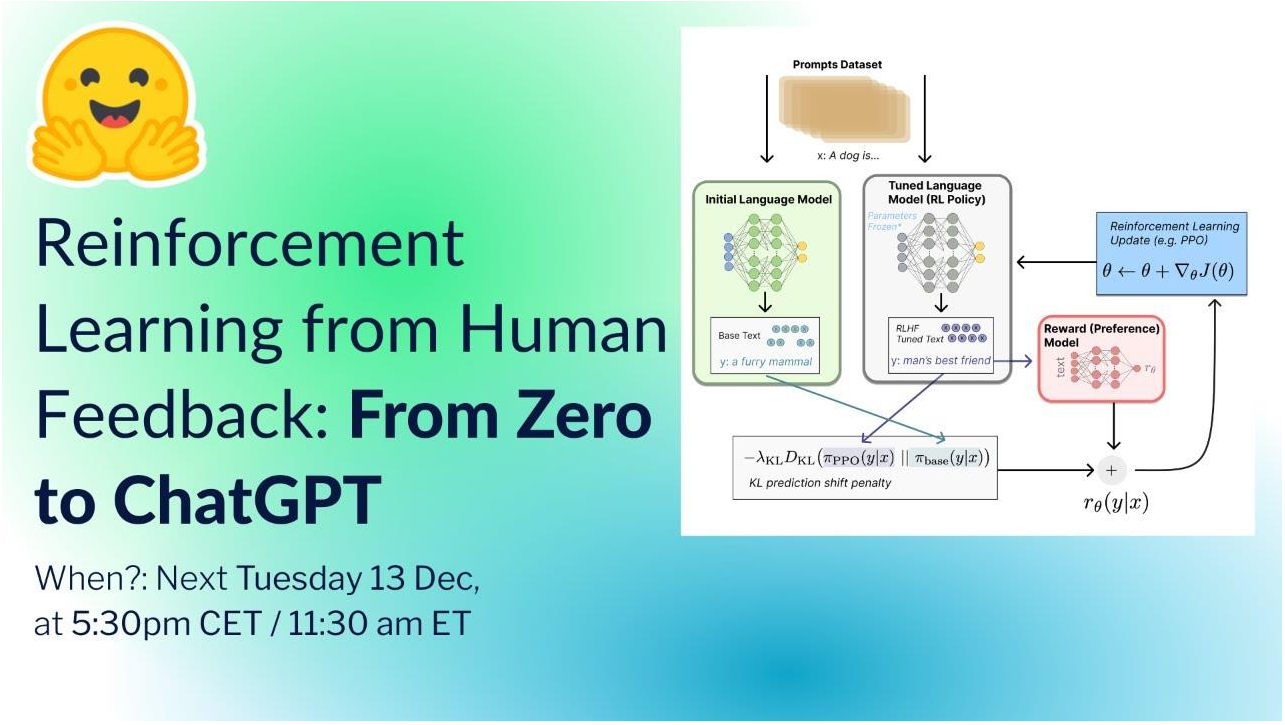
Update on GitHub


natolambert
Nathan Lambert


lcastricato
Louis Castricato guest


lvwerra
Leandro von Werra


Dahoas1
Alex Havrilla guest





Nathan Lambert

RLHF and Active Learning

	With Teacher	Without Teacher
Active	Reinforcement Learning / Active Learning	Intrinsic Motivation / Exploration
Passive	Supervised Learning	Unsupervised Learning

https://cnvrg.io/reinforcement-learning/



Questions and Challenges....

Questions and Challenges

- Can RLHF scale down to smaller budgets?
- How expensive is policy modification in light of new data?
- Do we really need reinforcement learning?
- How can we go beyond chat and information retrieval?
- **Can we improve the quality of outputs?**
- **How do we put these models to work?**
- **What is the long-term impact of all this?**



Improving the Quality of Outputs

LLMs on LSD*

REUTERS® World Business Legal Markets Breakingviews Technology Investigatic

Technology

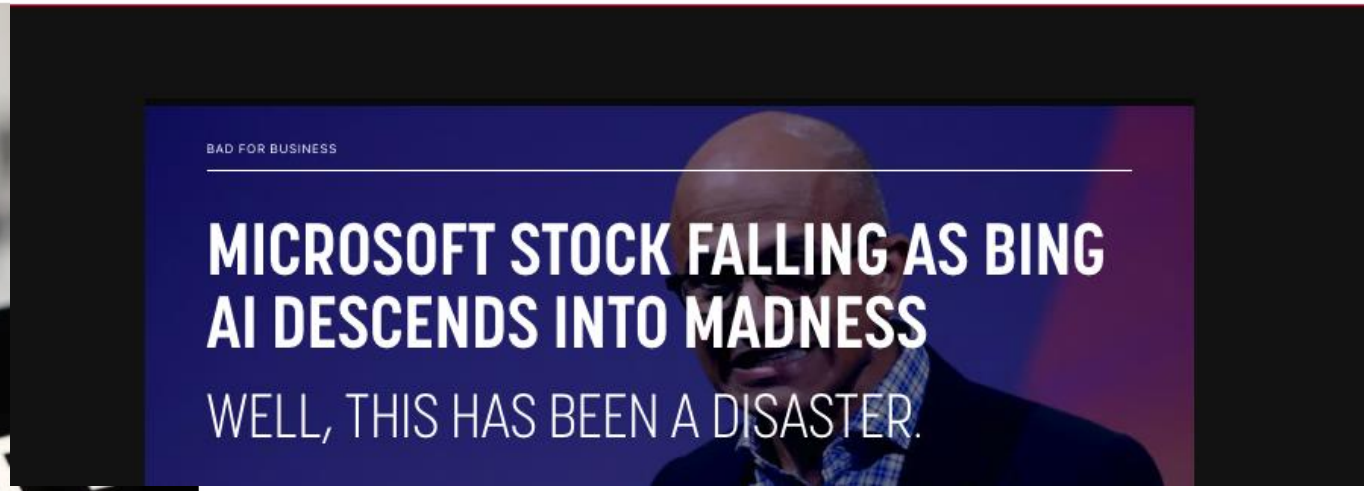
2 minute read · February 11, 2023 12:17 AM GMT · Last Updated 9 days ago

Google cautions against 'hallucinating' chatbots, report says

Reuters



Feb 17th THE BYTE.



*LSD = Drug which causes hallucinations

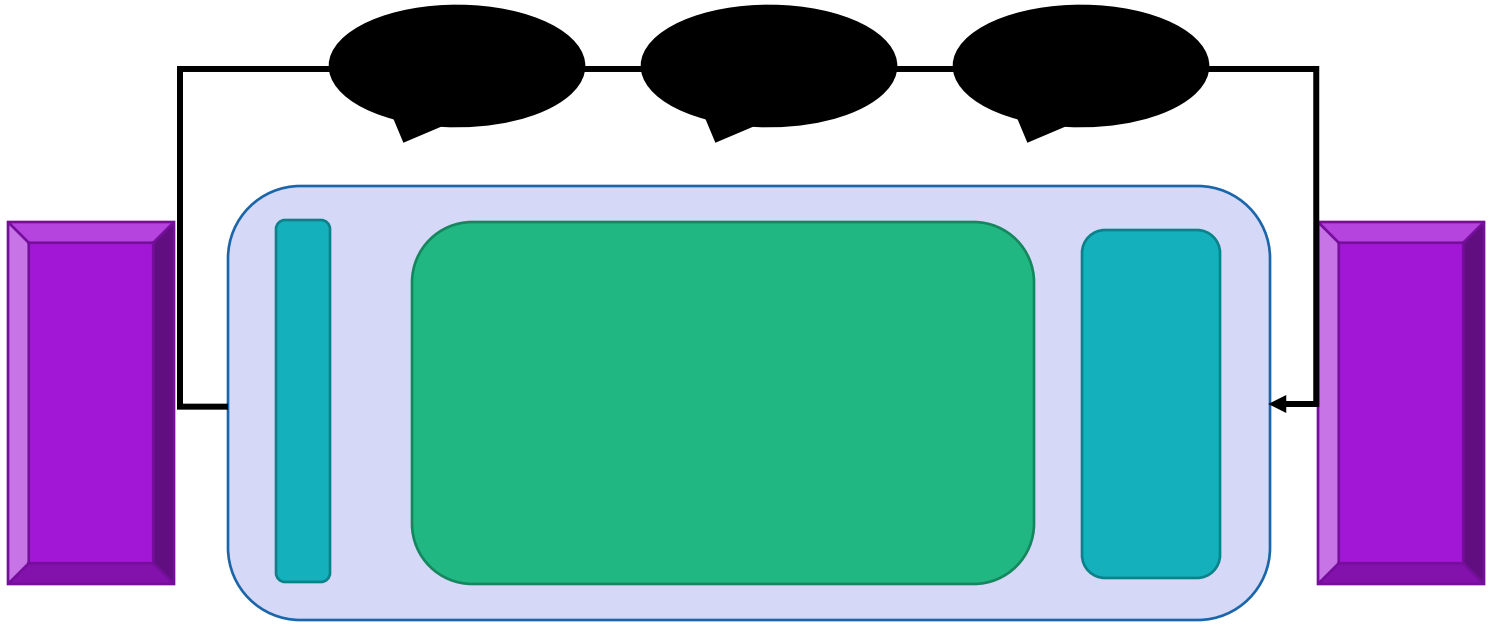


Dog catching a frisbee

Dog catching a Ball

Dog jumping in a field

Aligning the LLMs



Pre-Filter
User Input

LLM Core

Post-Filter
System Output

Domain Specific Examples

Moralistic and ethical language



Training and Using Large LLMs

LLM Usage Modes

- LLM as an API
  - Prompt Engineering
  - Smart Application Development

finxter **ChatGPT Prompting Cheat Sheet** OpenAI

How Prompt Engineering Works

Note: Prompt engineering designs and optimizes prompts for language models.

It's important in NLP and language generation. Prompt formats guide the model and can be used for tasks like product descriptions or conversational AI.

Reliable prompt formats exist, but exploring new formats is encouraged.

"{your input here}" is a placeholder for text or context

Rules of Thumb and Examples

***Rule #1** – Instructions at beginning and ### or """ to separate instructions or context*

✗
```
Rewrite the text below in more engaging language.

{your input here}
```

✓
```
Rewrite the text below in more engaging language.

Text: """
{your input here}
"""
```

***Rule #2** – Be specific and detailed about the desired context, outcome, length, format, and style.*

✗
```
Write a short story for kids
```

✓
```
Write a funny soccer story for kids that teaches the
kid that persistence is key for success in the style of
Rowling.
```

***Rule #3** – Give examples of desired output format*

✗
```
Extract house pricing data from the following text.

Text: """
{your text containing pricing data}
"""
```

✓
```
Extract house pricing data from the following text.

Desired format: """
House 1 | $1,000,000 | 100 sqm
House 2 | $500,000 | 90 sqm
... (and so on)
"""

Text: """
{your text containing pricing data}
"""
```

***Rule #4** – First try without examples, then try giving some examples.*

✓
```
Extract brand names from the text below.

Text: {your text here}

Brand names:
```

✓
```
Extract brand names from the texts below.

Text 1: Finxter and YouTube are tech companies. Google
is too.
Brand names 2: Finxter, YouTube, Google
###

Text 2: If you like tech, you'll love Finxter!
Brand names 2: Finxter
###

Text 3: {your text here}
Brand names 3:
```

***Rule #5** – Fine-tune if Rule #4 doesn't work*

Fine-tuning improves model performance by training on more examples, resulting in higher quality results, token savings, and lower latency requests.

GPT-3 can intuitively generate plausible completions from few examples, known as **few-shot learning.**

Fine-tuning achieves better results on various tasks without requiring examples in the prompt, saving costs and enabling lower-latency requests.

Example Training Data
```
{"prompt":"<input>", "completion":"<ideal output>"}
{"prompt":"<input>", "completion":"<ideal output>"}
{"prompt":"<input>", "completion":"<ideal output>"}
...
```

***Rule #6** – Be specific. Omit needless words.*

✗
```
ChatGPT, write a sales page for my company selling sand
in the desert, please write only a few sentences,
nothing long and complex
```

✓
```
Write a 5-sentence sales page, sell sand in the desert.
```

***Rule #7** – Use leading words to nudge the model towards a pattern*

✗
```
Write a Python function that plots my net worth over 10
years for different inputs on the initial investment
and a given ROI
```

✓
```
# Python function that plots net worth over 10
# years for different inputs on the initial
# investment and a given ROI

import matplotlib

def plot_net_worth(initial, roi):
```

***Bonus Prompt** – Let ChatGPT Design the Optimal Prompt*

```
New models are created frequently, and the performance of subsequent models can be an order of
magnitude. You are a robot for creating prompts. You need to gather information about the user's
goals, examples of preferred output, and any other relevant contextual information.

The prompt should contain all the necessary information provided to you. Ask the user more
questions until you are sure you can create an optimal prompt.

Your answer should be clearly formatted and optimized for ChatGPT interactions. Be sure to start
by asking the user about the goals, the desired outcome, and any additional information you may
need.
```



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Prompt Engineering for ChatGPT

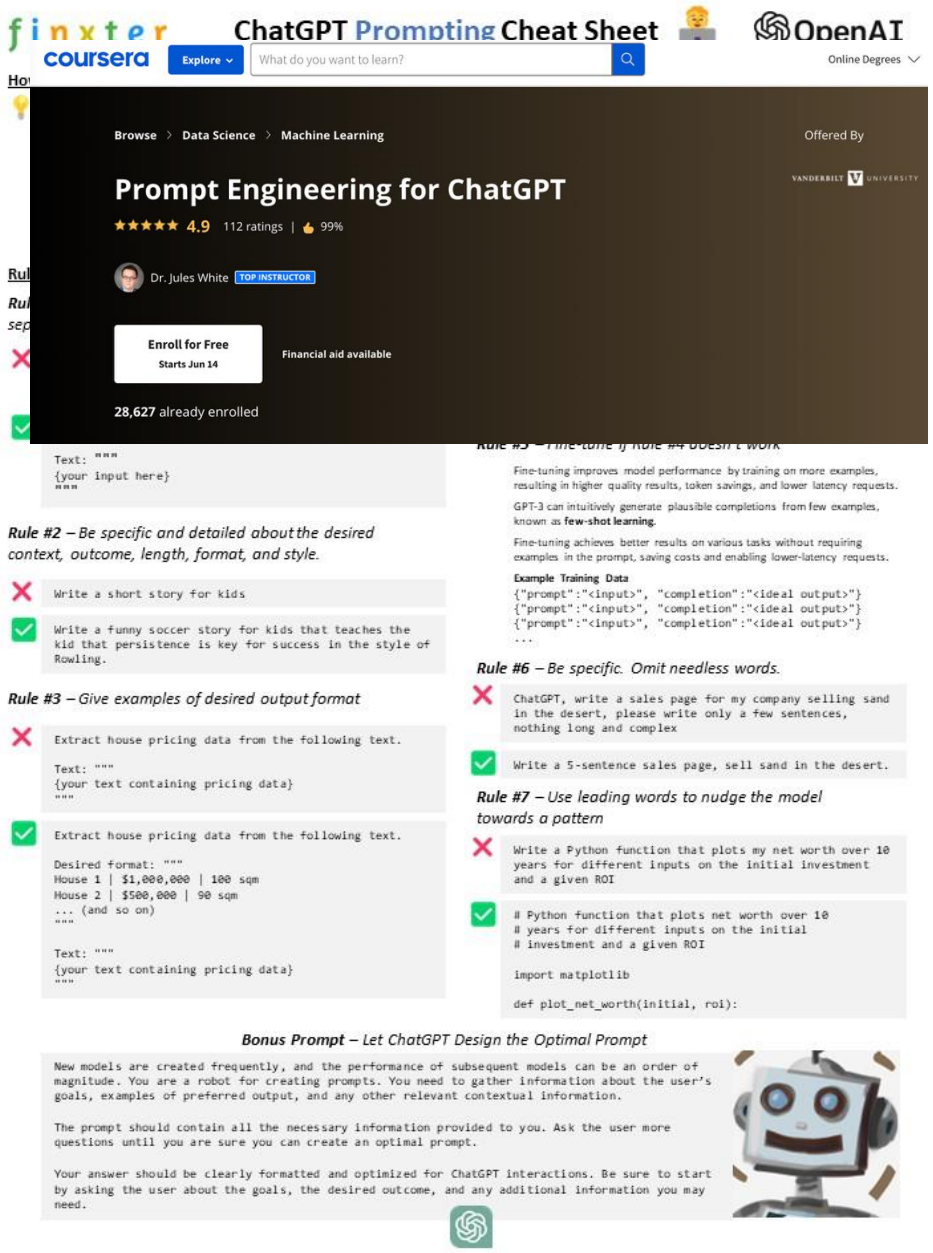
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- Prompt Engineering - Introduction
  - Prompt Engineering - LLM Settings
  - Prompt Engineering - Basics of Prompting
  - Prompt Engineering - Prompt Elements
  - Prompt Engineering - General Tips for Designing Prompts
  - Prompt Engineering - Examples of Prompts
- Prompt Engineering - Techniques
  - Prompt Engineering - Zero-Shot Prompting
  - Prompt Engineering - Few-Shot Prompting
  - Prompt Engineering - Chain-of-Thought Prompting
  - Prompt Engineering - Self-Consistency
  - Prompt Engineering - Generate Knowledge Prompting
  - Prompt Engineering - Tree of Thoughts (ToT)
  - Prompt Engineering - Automatic Reasoning and Tool-use (ART)
  - Prompt Engineering - Automatic Prompt Engineer
  - Prompt Engineering - Active-Prompt
  - Prompt Engineering - Directional Stimulus Prompting
  - Prompt Engineering - ReAct Prompting
  - Prompt Engineering - Multimodal CoT Prompting
  - Prompt Engineering - Graph Prompting
- Prompt Engineering - Applications
  - Prompt Engineering - Program-Aided Language Models
  - Prompt Engineering - Generating Data

LLM Usage Modes

- LLM as an API
  - Prompt Engineering
  - Smart Application Development
    - Few Shot vs. Zero Shot Prompting
    - Chain of Thought Prompting
    - Decomposition
  - LangChain
- Fine-tuning LLM
  - **We** can't fine-tune the proprietary LLMs
  - Is the Future Open?

An Open LLM Future?

a BigScience initiative

BLOOM

176B params · 59 languages · Open-access

MISTRAL AI_



🤗 Open LLM Leaderboard

Rank	Type	Model	Average	IFEval	BBH	MATH	GPQA
1	🔶	MaziyarPanahi/calme-3.2-instruct-78b	52.02 %	80.63 %	62.61 %	39.95 %	20.36 %
2	💬	dfurman/CalmeRys-78B-Orpo-v0.1	51.24 %	81.63 %	61.92 %	40.71 %	20.02 %
3	💬	MaziyarPanahi/calme-3.1-instruct-78b	51.20 %	81.36 %	62.41 %	38.75 %	19.46 %
4	💬	MaziyarPanahi/calme-2.4-rys-78b	50.71 %	80.11 %	62.16 %	40.41 %	20.36 %
5	🔶	newsbang/Homer-v1.0-Qwen2.5-72B	47.35 %	76.28 %	62.27 %	48.34 %	22.15 %
6	🔶	Sakalti/ultiima-72B	46.58 %	71.40 %	61.10 %	52.42 %	21.92 %
7	🔶	shuttleai/shuttle-3	45.99 %	81.54 %	64.05 %	41.69 %	21.59 %
8	🤝	rombodawg/Rombos-LLM-V2.5-Qwen-72b	45.91 %	71.55 %	61.27 %	50.68 %	19.80 %
9	🔶	zetasepic/Qwen2.5-72B-Instruct-abliterated	45.29 %	71.53 %	59.91 %	46.15 %	20.92 %
10	🔶	dnhkng/RYS-XLarge	45.13 %	79.96 %	58.77 %	41.24 %	17.90 %
11	🔶	raphgg/test-2.5-72B	45.03 %	84.37 %	62.15 %	30.82 %	18.57 %

LLMs and the Future of Work



ICT and the service economy

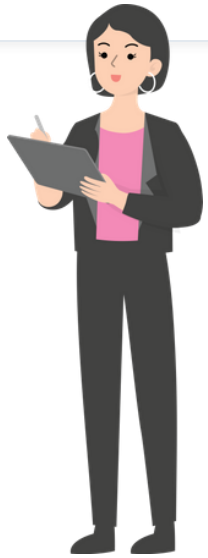


Will Robots
1.3M views • 1 m
CNBC
This robot named
CC



Where Hav
61K views • 8 da
Two Cents
QR codes, digita
CC

Impact on Knowledge Economy



Managers



Creatives



The Professions

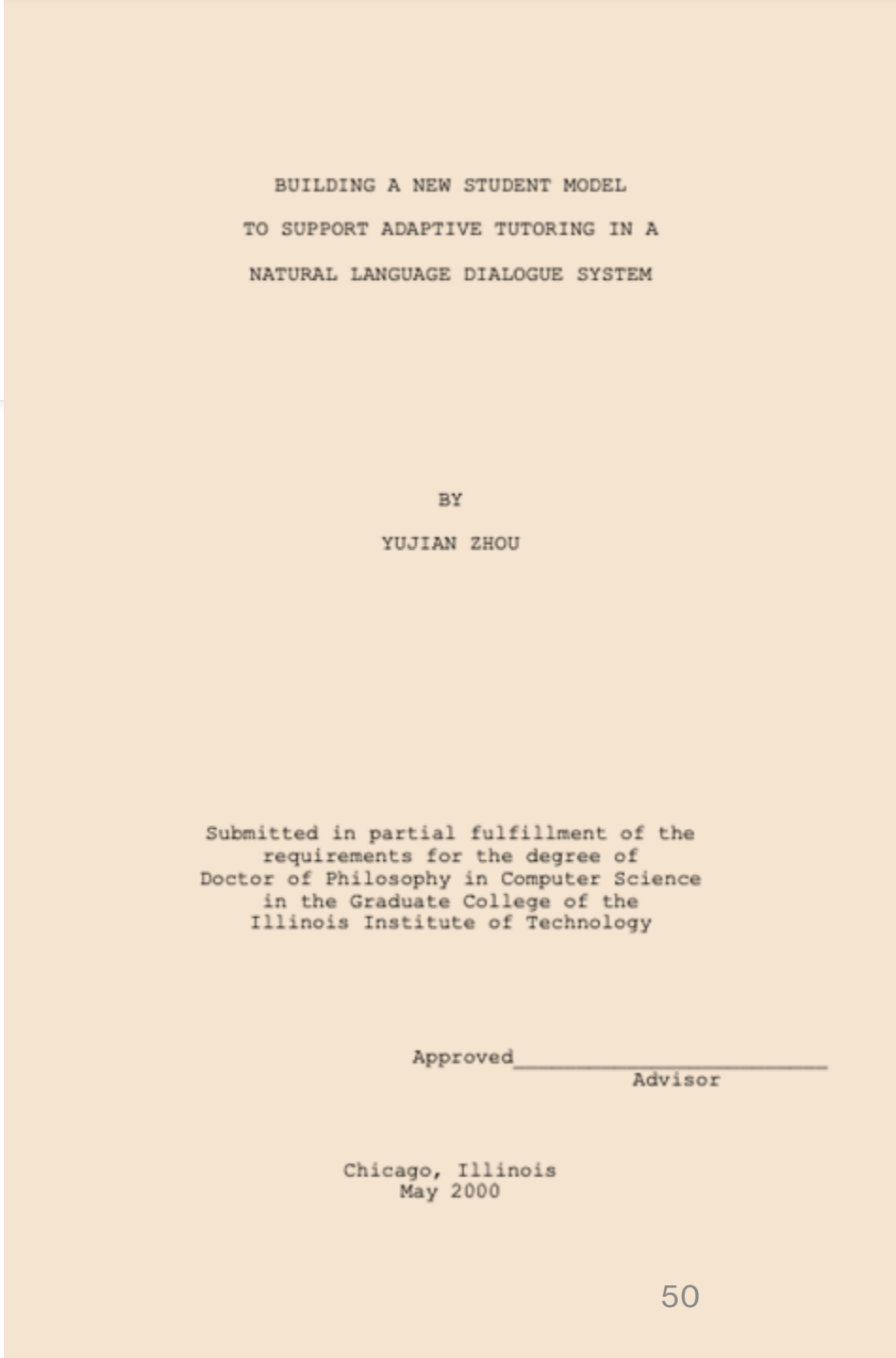


Engineers

Remember the Positives

- Democratization of Knowledge and Tutorage

- These tools have been the goal for many in educational AI for years!



BUILDING A NEW STUDENT MODEL

TO SUPPORT ADAPTIVE TUTORING IN A

NATURAL LANGUAGE DIALOGUE SYSTEM

BY

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Submitted in partial fulfillment of the
requirements for the degree of
Doctor of Philosophy in Computer Science
in the Graduate College of the
Illinois Institute of Technology

Approved_____
Advisor

Chicago, Illinois
May 2000